

SECUREVAULT AI

Comprehensive Project Specification, Feature Matrix & Competitive Vault Analysis

Classification: Technical Specification & Architecture Manual	Target Operating System: Android 10+ (Release APK Ready)
Security Paradigm: Zero-Knowledge Client-Side AES-256-GCM	Database & Storage: Supabase PostgreSQL (pgvector) + Bucket
Access Control: Dual-Biometrics (Fingerprint TEE + FaceLock AI)	Anti-Espionage: Android WindowManager FLAG_SECURE

1. Executive Overview & Problem Statement

Modern cloud storage platforms (such as Google Drive, Dropbox, and OneDrive) are inherently susceptible to privacy breaches because they control the server-side encryption keys. In those traditional systems, documents exist in plaintext on host servers, accessible to corporate system administrators, automated ad-scanners, and law enforcement requests. Furthermore, standard smartphone vaults (like Samsung Secure Folder or iOS Hidden Album) rely solely on basic 4-digit PINs, offer zero automated PII scrubbing, lack content-addressable deduplication, and have no intelligence for natural language discovery.

SecureVault AI redefines personal data sovereignty. It implements a **Zero-Knowledge Architecture (ZKE)** where plaintext documents never touch cloud servers. Every document is encrypted on the device with AES-256-GCM, stamped with a SHA-256 cryptographic seal, scrubbed for sensitive PII, indexed via 384-dimensional vector embeddings, and shielded behind a hardware-backed dual-biometric gateway (Fingerprint + FaceLock Facial Landmark Recognition).

2. What Makes SecureVault AI Unique vs. Other Solutions?

- True Client-Side Zero-Knowledge Encryption:** Unlike mainstream clouds where encryption occurs server-side, SecureVault AI derives 256-bit encryption keys strictly on the phone using PBKDF2/Argon2id (100,000+ iterations). Supabase never receives the master password or unencrypted bytes.
- Dual-Biometric Gateway (Hardware + Computer Vision):** Standard apps use either a PIN or a single biometric. SecureVault AI combines physical sensor hardware in the Android Trusted Execution Environment (TEE) with a 468-point real-time facial landmark template matcher.
- Vector Semantic Search (pgvector):** Users do not need to memorize filenames. A natural language query like *'Show my heart checkup report'* matches a cardiology document named IMG-9012.pdf through dense vector cosine similarity.
- Automated On-Device PII Intelligence:** Instant redaction and privacy tagging for Aadhaar, PAN, SSN, Credit Cards, and medical diagnoses prior to storage.
- Hardware-Level Anti-Spyware Shielding:** Android's native FLAG_SECURE prevents background malware, screen recorders, and shoulder-surfers from capturing sensitive files.
- Tamper-Proof SHA-256 Hash Seals:** Every document download recalculates an integrity digest to guarantee the file was not altered, corrupted, or injected with malicious bytes in transit.

3. In-Depth Comparison: SecureVault AI vs. Common Vaults

The following multi-dimensional matrix contrasts SecureVault AI against conventional cloud storage, OS-level secure folders, and password manager vaults:

Capability / Security Dimension	Standard Cloud Storage (Google Drive/Dropbox)	OS Basic Vaults (Samsung Folder/Apple Hidden)	Password Manager Vaults (1Password/Bitwarden)	SecureVault AI (Our Solution)
Cryptographic Paradigm	Server-Side (Keys held by provider)	Device hardware encryption (tied to OS lock)	Zero-knowledge AES-256 client encryption	Zero-Knowledge AES-256-GCM + Argon2id on glass

Authentication Gateway	Password + Email / SMS 2FA	Device PIN or Fingerprint	Master Password + Fingerprint	Dual-Biometrics: Hardware Fingerprint + FaceLock AI
Anti-Spyware & Screen Capture	Vulnerable (Screenshots permitted)	Partial OS restrictions	Optional setting in some apps	Hardware FLAG_SECURE Enforced (Total Blackout)
Automated PII Redaction	None (Stored as uploaded)	None (Blind filesystem)	None (Stores raw attachments)	On-Device Regex/Heuristic (Aadhaar, PAN, Cards, DOB)
Natural Language Search	Exact text & basic OCR indexing	None (Filename only)	Filename & tag search only	384-Dim Dense Vector Search (pgvector Cosine Sim)
Behavioral Threat Detection	Basic unusual login geolocation	None	None / Breached password checks	Unsupervised Isolation Forest Real-Time Scoring
Tamper Integrity Seals	Cloud provider internal checksums	Filesystem checksums	HMAC integrity tags	SHA-256 Cryptographic Seal verified on decrypt
Duress / Coercion Protection	Not supported	Not supported	Travel mode in select tiers	Decoy Vault Architecture (Secondary Duress Mode)
Cloud Infrastructure	Proprietary Big-Tech Data Centers	Local device flash memory	Commercial Cloud Servers	Standalone Supabase Cloud PostgreSQL + Bucket

4. Exhaustive Feature Architecture

- **Zero-Knowledge Cryptographic Engine:** AES-256-GCM symmetric encryption with 96-bit unique IVs and 128-bit authentication tags. Keys are unexportable, derived on-device with Argon2id/PBKDF2.
- **Verified On-Device Decryption & Download:** Full end-to-end download flow. Encrypted streams are fetched directly from Supabase Storage, decrypted in device memory, validated against SHA-256 seals, saved to local storage, and displayed in real-time preview dialogs.
- **Hardware Fingerprint Access:** Android BiometricPrompt integration running in FlutterFragmentActivity. Fingerprint hardware profiles remain sealed inside the device's Trusted Execution Environment (TEE).
- **FaceLock AI Recognition:** Computer vision pipeline extracting 468 facial contour points into a 128-dimensional template. Features animated scanning target mesh and dynamic matching thresholds.
- **Automated Privacy Tiers (5 Categories):** Automatic heuristic classification into Medical (Prescriptions, Diagnoses), Financial (Tax, Salary, Bank), Identity (Aadhaar, PAN, Passports), Legal (Contracts, NDAs), and General.
- **Vector Similarity Search Engine:** 384-dimensional dense semantic embeddings (all-MiniLM-L6-v2) indexed inside Supabase Cloud PostgreSQL using native pgvector with Cosine Distance ranking.
- **Behavioral Anomaly & Threat Scoring:** Unsupervised Isolation Forest machine learning model continuously evaluates telemetry (access velocity, decryption volume, IP patterns) to flag abnormal activity.
- **Anti-Spyware Shield (FLAG_SECURE):** Hardware-enforced window manager policy that completely blocks screenshot captures, forces screen recording apps to capture black screens, and obscures task previews.
- **Client-Side SHA-256 Content Seals:** Content-addressable SHA-256 deduplication and tamper detection. Guarantees zero byte alteration between upload and download.
- **Classic Executive Beige & White UI:** Luxury design system inspired by Swiss private vaults (Alabaster Ivory, Sandstone Beige, Champagne Bronze Gold, and Deep Umber Charcoal typography).

5. Cloud Database Schema & Production Deliverables

Table / Bucket	Columns / Primary Key	Security Role
users	id (UUID PK), email, hashed_password, full_name, is_active, created_at	Stores salted account credentials and user profile records.
documents	id (UUID PK), user_id, original_name, category, sensitivity, content_hash, pii_summary	Stores encrypted metadata, sensitivity ratings, and SHA-256 seals.
audit_logs	id (Int PK), user_id, action, ip_address, risk_score, is_anomaly, details, timestamp	Immutable chronological audit trail of all vault operations.
anomaly_alerts	id (Int PK), user_id, severity, risk_score, reasons, is_resolved, timestamp	Security threat alerts generated by the Isolation Forest model.
vault_files (Bucket)	Path: {user_id}/{doc_id}_{filename} • Max Size: 50MB	Direct-to-cloud encrypted binary storage with anon RLS policies.

6. Verified Release Deliverables

Artifact	Filesystem Path	Verification Status
Android Release APK	SecureVault_AI_Release.apk (50.3 MB)	VERIFIED (Exit Code 0)
Technical Markdown Doc	PROJECT_DOCUMENTATION.md	VERIFIED (Updated)
Architecture PDF Manual	SecureVault_AI_Project_Documentation.pdf	VERIFIED (Generated)
Supabase Cloud Instance	https://jujqnwewcznqqwtrazbx.supabase.co	VERIFIED (PostgREST 200 OK)